Low-Rank Adaption on Transformer-Based Oriented Object Detector for Satellite Onboard Processing of Remote Sensing Images

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***Abstract*— Deep learning models in satellite onboard enable real-time interpretation of remote sensing images, reducing the need for data transmission to the ground and conserving communication resources. As satellite numbers and observation frequencies increase, the demand for satellite onboard real-time image interpretation grows, highlighting the expanding importance and development of this technology. However, updating the extensive parameters of models deployed on the satellites for spaceborne object detection model is challenging due to the limitations of uplink bandwidth in wireless satellite communications. To address this issue, this article proposes a method based on parameter-efficient fine-tuning (PEFT) technology with a low-rank adaptation module. It involves training low-rank matrix parameters and integrating them with the original model's weight matrix through multiplication and summation, thereby fine-tuning the model parameters to adapt to new data distributions with minimal weight updates. The proposed method combines PEFT with full fine-tuning in the parameter update strategy of the oriented object detection algorithm architecture. This strategy enables model performance improvements close to full fine-tuning effects with minimal parameter updates. In addition, the low-rank approximation is conducted to explore the intrinsic dimensions of parameter matrices in normal-size models. Extensive experiments conducted on the DOTAv1.0, HRSC2016, and DIOR-R datasets verify the effectiveness of the proposed method. By fine-tuning and updating only 12.4% of the model's total parameters, it is able to achieve 97%–100% of the performance of full fine-tuning models. In addition, the reduced number of trainable parameters accelerates model training iterations and enhances the generalization and robustness of the oriented object detection model. The source code is available at: https://github.com/fudanxu/LoRA-Det.**

***Index Terms*— Low-rank adaption (LoRA), oriented object detection, parameter efficient fine-tuning (PEFT).**

I. Introduction

AS ARTIFICIAL intelligence technology advances, object detection models based on deep learning are playing an increasingly important role in the task of remote sensing image interpretation. The current approach typically involves transmitting images captured by satellites to ground stations, where deep learning models then perform inference and interpretation. However, processing massive volumes of remote sensing images through ground station models places a significant burden on satellite communication transmission and results in noticeable interpretation delays. This issue is exacerbated by the rapid increase in the number of satellites and the enhanced frequency of remote sensing observations, supported by advancements in aerospace technology. Therefore, satellite onboard models provide a reasonable potential for effective remote sensing image interpretation by directly processing the real-time captured images on each satellite without image transmission.

Moving forward, it is anticipated that onboard satellite deep learning models will be frequently updated to enhance both performance and generalization after retraining with continuous data streams. However, the full updating of all model parameters presents a considerable challenge and is time-intensive due to the limited uplink bandwidth available and low transmission rates in satellite communications. Fig. 1 highlights the constraints on uplink transmission rates for spaceborne models relative to the more flexible downlink rates, showing a limited uplink rate of 64 kbps for command and control versus a high-capacity downlink rate of 520 Mbps for science data, reflecting the disparity between the bandwidth needs of uplink and downlink in Earth observation satellite systems like Sentinel-2.

Consequently, we aim to develop a sophisticated algorithm that necessitates updating only a subset of the model's parameters to achieve enhanced detection performance. After retraining the detection model with an abundance of new remote sensing image data on a ground platform, only this subset of parameters will be modified and scheduled for upload to the onboard satellite model via uplink communication. This method is specifically designed to boost model performance while simultaneously minimizing the bandwidth requirements for uplink transmissions in satellite communications.

Inspired by parameter-efficient fine-tuning (PEFT) for large models, which enables training and updating only a small fraction of parameters, we leverage PEFT to enhance model performance and approximate full fine-tuning effects under resource constraints. Widely used in foundation models like large language models (LLMs) [1] to reduce training costs, PEFT is still in the early stages for vision models. Some

Received 8 November 2024; accepted 29 December 2024. Date of publication 2 January 2025; date of current version 16 January 2025. This work was supported by the Science and Technology Commission of Shanghai Municipality under Grant 23JC1400501. *(Corresponding author: Feng Xu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3524578

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Fig. 1. Comparison of uplink and downlink transmission limitations for spaceborne object detection models. *Specific transmission rate parameters are sourced from the official documentation of the low Earth orbit remote sensing satellite, Sentinel-2.

applications, such as segmentation with the segment anything model (SAM) [2] and generation with diffusion models [3], have begun to adopt PEFT.

To address the challenge of satellite onboard model parameter updating, we intend to adopt the low-rank adaption (LoRA) module [4] to fine-tune the satellite onboard object detection model as a PEFT strategy. We use continuously incoming new remote sensing data, employing the minimal volume of trainable parameters possible without altering the model structure. Furthermore, only these trainable parameters that require updating are transmitted and then integrated into the weights of the object detection model onboard the satellite payload. This explains the use of PEFT techniques for training oriented object detection models in remote sensing images. The aim is to enhance the detection capabilities and generalizability of these models.

The existing investigation and development of PEFT technology in object detection algorithms is not sufficient [5]. This is due to the lack of a foundational model in object detection and the complexity of detection algorithm architectures. Most PEFT methods that have been extensively studied are applied to the backbone of deep learning networks. However, there is a lack of investigation concerning the full scope of the algorithm architecture for diverse downstream tasks.

Furthermore, PEFT module designs of network structure mostly focus on common LLMs with transformer frameworks [6], where backbone weights constitute the majority of the parameters. This has led to prevalent PEFT algorithms being primarily explored within the attention and multilayer perceptron (MLP) modules of a transformer-based backbone. Similarly, the LoRA method is employed on the transformer module for fine-tuning stable diffusion models [3]. This approach manipulates representations in the latent space to accommodate a variety of image-generation applications. Nevertheless, there is still limited research on PEFT techniques for object detection models based on convolutional neural networks. Specifically, there is a gap in studies focusing on fine-tuning parts beyond the backbone structures.

In summary, we present three challenges of PEFT application on object detection tasks.

1) Object detectors have more complicated architectures and specially-designed modules, compared with image-level classification and pixel-level segmentation. Specifically, the hierarchical feature map and region proposal generation add complexity to the design of PEFT methods. In addition, the fine-grained detection head blocks present further challenges.
2) For spaceborne onboard processing scenarios, satellite hardware imposes limitations on the parameter quantities and computing resources of the detectors. Consequently, we are interested in the application of PEFT on normal-sized models instead of large models, which contain a few millions of parameters [7]. It is found that some empirical principles of PEFT on large models are no longer feasible for these models.
3) Selecting a proper low-rank value for trainable matrices in the LoRA module on normal-sized models is both critical and challenging. This is because the intrinsic rank may be relatively high, which differs from the hypothesis that matrices in large models tend to contain low intrinsic dimensionality [8].

To tackle these challenges, this article explores the architectural design of PEFT techniques in the context of oriented object detection algorithms. It involves with identifying the layers that are most important for fine-tuning and determining the optimal structures for hybrid PEFT methods to leverage new data in enhancing the model's detection capabilities. In addition, it includes figuring out a reasonable low-rank value for each matrix in LoRA modules to optimize the tradeoff between model performance and resource usage.

The main contributions of this article can be summarized as follows.

1) The proposed method designs a novel parameter efficient fine-tuning algorithm for oriented object detection named LoRA-Det, conducting LoRA module through the whole framework including transformer-based backbone and detection head.
2) The necessity and effective strategies for adopting PEFT in comprehensive modules of the detection network are explored. A hybrid PEFT and full fine-tuning strategy in transformer and convolutional layers is proposed to achieve a satisfactory tradeoff between detection performance and the quantity of trainable parameters.
3) A low-rank approximation method is employed to analyze the intrinsic dimensions of the trainable matrices for determining the rank r value in LoRA modules. Extensive experiments are conducted on multiple remote sensing image datasets to verify the effectiveness and rationality of the proposed algorithm.

The rest of this article is arranged as follows. Section II presents an extensive review of prior studies concerning LoRA-Det. Section III elaborates on the detailed architecture of the proposed method LoRA-Det. Section IV presents a thorough analysis of quantitative and qualitative evaluation

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results, comparing multiple experiments and conducting an ablation study. Finally, Section V concludes the article.

II. Related Works

A. Parameter Efficient Fine-Tuning

Parameter-efficient fine-tuning technologies are widely used in adapting large models to diverse downstream tasks with minimal additional parameters and computational resources. Initially developed for LLMs in natural language processing, PEFT has since been applied in computer vision, including vision transformers (ViTs) and diffusion models.

Serial adapter [9] introduces additional trainable parameters by inserting two lightweight adapter layers into the transformer blocks of original networks, one after the self-attention layer and the other after the FFN layer. AdapterFusion [10] improves the computation efficiency by adding adapter layers after "Add & Norm." Parallel adapter (PA) [11] proposes a parallel side-network structure of adapters on transformer blocks, similarly as CIAT [12] and KronA [13]. Such adapter-based methods add extra layers to original model architectures and some of them conduct multitask learning such as AdaMix [14] and AdapterSoup [15].

Another way of PEFT based on LoRA introduces no inference latency for the model by updating existing parameters through matrix multiplication. LoRA is the most widely recognized reparameterization technique of PEFT. DyLoRA [16] proposes a strategy to select the proper rank for the LoRA module and reduce the time cost of the training process. AdaLoRA [17] leverages singular value decomposition (SVD) to obtain three weight matrices of the LoRA module for fine-tuning. Multiple research is focused on improving the performance of LoRA via diverse approaches including Laplace approximation [18] and novel training strategies [19]. DoRA [20] designs weight-decomposed low-rank adaptation by decomposing model weights into magnitude and direction and introduces a novel fine-tuning strategy for weight matrices.

B. LoRA in Computer Vision

The application of PEFT technology in various vision tasks and vision-language models [21], [22] has also been widely investigated. Considering LoRA focus on fine-tuning the Q, K, and V matrices of parameter weight in attention modules, this approach is mainly explored as a fine-tuning strategy in the ViT [23] backbone of large models, such as SAM and diffusion models. LoRAT [24] leverages the LoRA module to fine-tune the transformer encoder of the ViT-g backbone and designs an MLP-based head to improve model performance and reduce training costs for visual object tracking. Lian et al. [25] delve into the LoRA method for PEFT technologies on medical vision foundation models. In [26], LoRA is utilized on a pretrained ViT backbone and single center loss (SCL) is conducted to enhance the generalization capability of the face forgery detection model. The introduction of concept sliders [27] presents plug-and-play LoRA adapters, facilitating precise adjustments to concepts (e.g., age, smiling) within a diffusion model. A novel semantic segmentation model is proposed by [28], fine-tuning the ViT-based image encoder of SAM with a low-rank adaptation approach for feature extraction on optical aerial images. AiRs [5] explores diverse PEFT technologies, such as LoRA, adapter, and bitfit, on foundation models and designs adapter-based modules for multiple remote sensing downstream tasks.

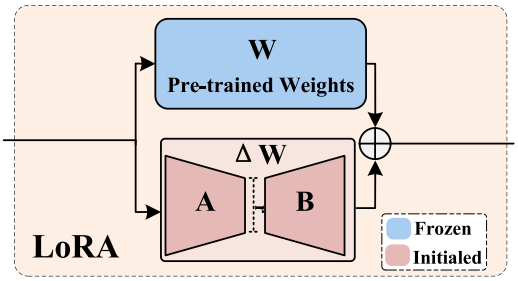


Fig. 2. Structure of LoRA module.

C. Oriented Object Detection

Oriented object detection, a popular topic in computer vision, has garnered comprehensive attention in both algorithmic advancements and practical applications. In response to the challenges posed by densely distributed objects with arbitrary orientations in remote sensing imagery, numerous algorithms have been investigated to accurately locate rotated bounding boxes encompassing object edges on publicly available aerial image datasets from satellites, such as DOTA [29], HRSC2016 [30], and DIOR-R [31], among others. The region of interest (RoI) transformer [32] incorporates the RRoI learner and rotated position-sensitive RoI align. The former learns the transformation from horizontal RoIs to rotated RoIs, while the latter extracts rotation-invariant features from RRoI to facilitate oriented object detection tasks. Subsequently, the rotation-equivariant detector (ReDet) [33] integrates rotation-equivariant networks into the backbone and utilizes rotation-invariant RoI align (RiRoI Align) to extract rotation-invariant features, leading to significant improvements in detection performance for aerial images with oriented bounding boxes. The oriented R-CNN [7] utilizes an oriented region proposal network (oriented RPN) and oriented R-CNN head to achieve state-of-the-art detection accuracy on two public datasets, enabling the acquisition of high-quality oriented proposals and refined oriented regions of interest. RTMDet [34] proposes an efficient light-weighted real-time object detector and is utilized on oriented object detection in aerial images.

III. Methodology

A. LoRA in LoRA-Det

1) Preliminaries of LoRA: Deep neural networks contain massive parameters for dense matrix multiplication. The parameter matrices in a well-pretrained weight tend to have full-rank, while updating parameters by training new data under a similar scene for an identical task, the updated matrices can be restricted in low dimension to decrease the number of updated parameters by LoRA [4].

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As shown in Fig. 2, to update a parameter matrix from $W \in \mathbb{R}^{d\times k}$ in the pretrained weight where $d \times k$ denotes the dimensions of the matrix W to $W' \in \mathbb{R}^{d\times k}$ as

$$W' = W + \Delta W \tag{1}$$

where $\Delta W \in \mathbb{R}^{d\times k}$, LoRA decomposes ΔW to two low dimension matrices A and B as

$$\Delta W = BA \tag{2}$$

where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$. By utilizing LoRA fine-tuning, the number of learning parameters is converted from $d \times k$ to $r \times (d + k)$. By adjusting the rank r, the number of learning parameters is reduced when $r < (d \times k)/(d + k)$, and the quantity of fine-tuned trainable parameters can be extremely reduced if rank $r \ll \min(d, k)$. Therefore, the feature extraction of forward propagation in the LoRA block can be described as

$$f = W'x = (W + \Delta W)x = (W + BA)x = Wx + BAx. \tag{3}$$

During the training process, the weight matrix W is frozen and does not require gradient calculation. Initialized matrix A follows random Gaussian distribution as $A = \mathcal{N}(0, \sigma^2)$ and matrix B is zero as $B = 0$, and these two matrices contain trainable parameters for gradient update and fine-tuning. Therefore, $\Delta W = BA$ is zero so that it will not affect the feature extraction at the beginning of model training. In addition, W and ΔW have the same matrix dimension and can be summed coordinate-wise directly as

$$W' = W + \Delta W = W + BA \tag{4}$$

so LoRA maintains the original framework unchanged and brings no extra inference cost and latency after weight integration.

2) LoRA in Dense Blocks: LoRA can be employed in diverse structures of dense layers in neural networks by simply replacing the dense matrices in the pretrained weights. The proposed method designs two types of LoRA-based PEFT modules for oriented object detection algorithms, including LoRA in linear layers and LoRA in Swin transformer blocks.

For linear layers, the low-rank matrices A and B are directly added in parallel next to the parameter matrix of the fully connected layer as illustrated in Fig. 2.

For Swin transformer blocks [35], the LoRA fine-tuning structure is depicted in Fig. 3. In two successive Swin transformer blocks, the image feature is partitioned evenly without overlapping into 8×8 patches and each 4×4 patch is regarded as a local window where the multihead self-attention is computed. As a result, 2×2 windows are built for multihead self-attention in a Swin transformer block. For the first Swin transformer block, the windows of size 4×4 patches are partitioned regularly from top-left to right-bottom, while in the second block, the start of window partitioning is shifted by $(2, 2)$ patches, and an efficient batch computation method named cyclic shifting is deployed to restore four full local windows and avoid additional attention computation from extra subwindows.

After the window partition, the multihead self-attention with relative position bias is calculated in each local window as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^{\mathsf{T}}}{\sqrt{d_{\text{head}}}} + B\right)V \tag{5}$$

where $Q, K, V \in \mathbb{R}^{M^2\times d}$ are the query, key, and value matrices; d_{head} is the query/key dimension; $B \in \mathbb{R}^{M^2\times M^2}$ and M^2 is the number of patches in a window.

We utilize the LoRA layer to adapt the parameter matrices W_q and W_v in the attention module of Swin transformer blocks following the empirical experiments in LoRA [4], and other layers in the transformer blocks are frozen during the fine-tuning process.

Therefore, by employing this approach, which leverages the LoRA module for fine-tuning the backbone, there will be a substantial reduction in the number of trainable parameters. Depending on the value of rank r, it is reasonable to adjust the amount of parameters necessitated for training. This enables the model to be flexibly adapted to new data domains.

Based on the above introduction, the LoRA module relies on the low-rank intrinsic dimension of the parameter matrix, which enables it to perform optimal parameter efficient fine-tuning. When the parameter matrix is in the multihead attention module or fully connected network, and has a large matrix dimension, a lower rank r of the LoRA module is suitable for use in low-rank adaptive algorithms, achieving great fine-tuning performance while training fewer parameters. Therefore, the Oriented R-CNN algorithm [7] with Swin transformer backbone and bounding box head composed of two shared fully connected layers is selected as the baseline of LoRA-Det.

B. Parameter Efficient Fine-Tuning in Oriented Object Detection

As an oriented object detection network shown in Fig. 4, the Oriented RCNN [7] is fundamentally comprised of the backbone based on Swin transformer, feature pyramid network (FPN), oriented region proposal network (RPN) head, and bounding box head. An analysis of the parameter quantity distribution, as depicted in Fig. 5, reveals that the backbone and detection head (named as bounding box head in Oriented RCNN) constitute the bulk of the parameters, serving the critical functions of feature extraction and detection regression and classification, respectively. Therefore, the LoRA technique is employed within both the backbone and bounding box head components as shown in Fig. 4. This strategy not only diminishes the volume of parameters necessitating training and expedites the training process but also leverages PEFT methodologies to substantially improve the model's detection capabilities across new data domains.

1) LoRA Fine-Tuning on Transformer-Based Backbone: In the Swin transformer backbone, LoRA is applied within the attention modules of the transformer blocks. As illustrated in Fig. 4, an input image, after undergoing a nonoverlapping patch partition, sequentially passes through four stages to extract image features. Stage 1 consists of a linear embedding and two Swin transformer blocks, while stages 2, 3, and

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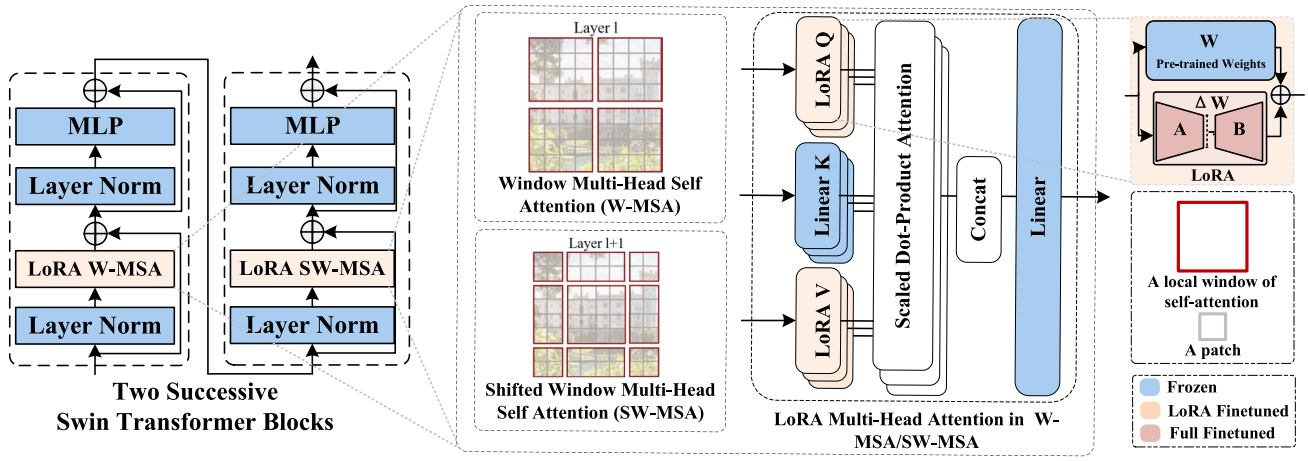


Fig. 3. Architecture of LoRA module in Swin transformer block.

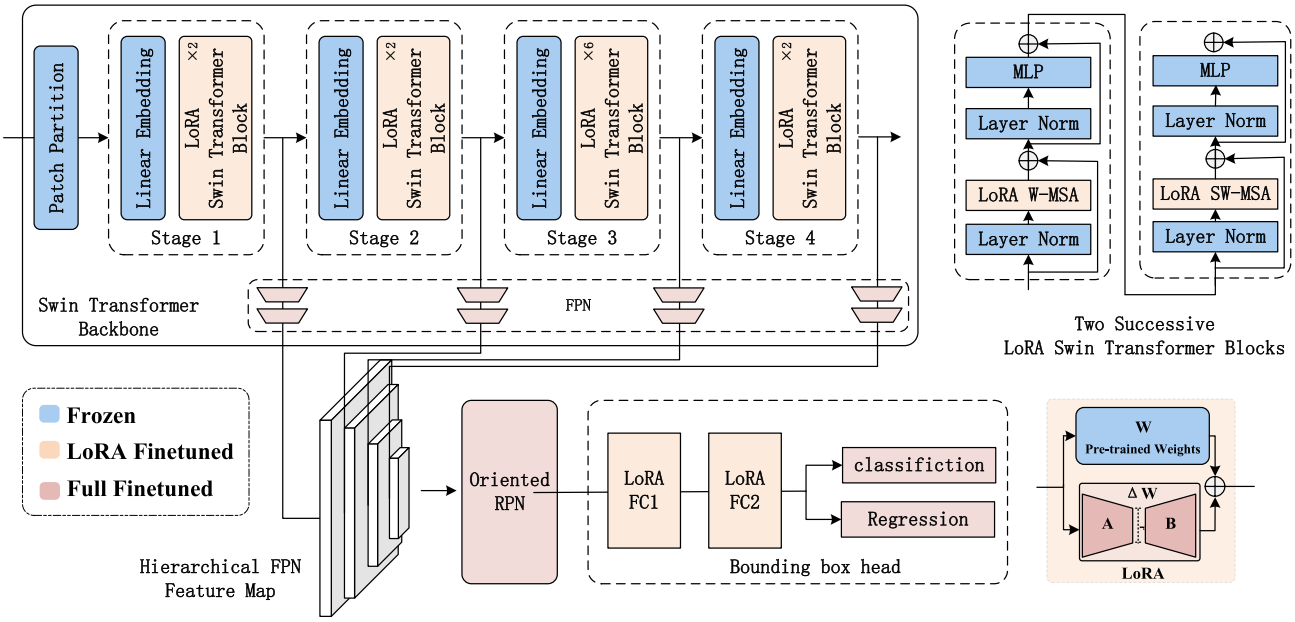


Fig. 4. Architecture of the proposed LoRA-Det.

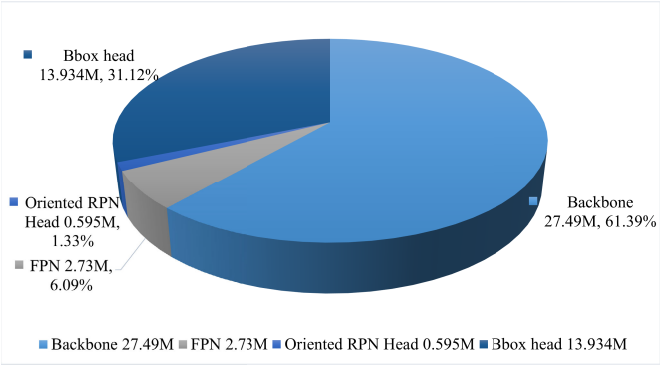


Fig. 5. Parameter distribution of oriented object detection model oriented RCNN.

4 each include a patch merging and multiple successive Swin transformer blocks. Specifically, stages 2, 3, and 4 are equipped with two, six, and two Swin transformer blocks, respectively. The structure of every two consecutive Swin transformer blocks, as shown in Fig. 3, integrates the LoRA module within the query and value matrices of the window multihead self-attention (W-MSA) and shifted window MSA (SW-MSA) modules following the strategies described in Section III-A.

2) Hybrid Fine-Tuning in Oriented RCNN: In the Oriented RCNN framework, the efficient fine-tuning of parameters in the detection head is realized through the adaptation of low-rank LoRA to the fully connected layers. After the Oriented RCNN head performs oriented RoI alignment on the oriented proposals, the resulting fixed-size feature vector is fed into the fully connected layers. LoRA module is applied to these two parameter-dense fully connected layers, utilizing low-rank matrices to fine-tune the shared fully connected layers of the detection head.

The proposed algorithm employs PEFT for the backbone and Oriented RCNN head using the LoRA module, as outlined earlier. The remaining components, primarily convolutional

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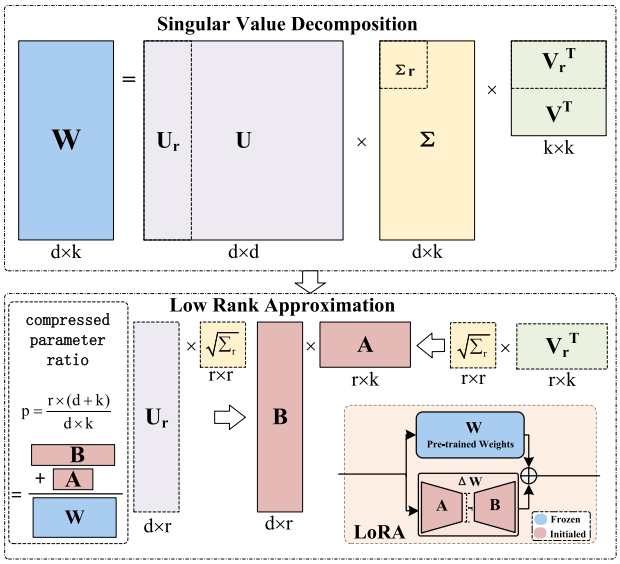


Fig. 6. Low-rank approximation method in LoRA-Det.

layers with fewer parameters, are subjected to standard full fine-tuning. The full fine-tuning approach encompasses modules such as the FPN and the Oriented RPN head.

Moreover, to augment the detection and classification performance of the model, the lightweight classification and regression fully connected layers located at the terminal part of the Oriented RCNN detection head also employ the full fine-tuning strategy. This method ensures a balanced optimization across the network, enhancing the model's overall performance by fully adjusting both its core and auxiliary components.

C. Low-Rank Approximation

To apply the low-rank adaptation algorithm in normal-sized satellite onboard detectors, we face challenges in identifying low-rank structures in each parameter matrix and selecting appropriate rank values r for LoRA modules to balance trainable parameters with fine-tuning performance. Thus, we explore low-rank approximation on LoRA-Det's parameter matrices. Inspired by SVD for data compression, we analyze the intrinsic rank and approximation errors of these matrices to determine optimal r values for the LoRA module, as shown in Fig. 6.

For a weight matrix $W \in \mathbb{R}^{d\times k}$ in deep learning network and its rank $r \leq \min(d, k)$, if the matrix is from large model with huge dimensions, its intrinsic rank might be low as depicted in [8], whereas the normal-size matrix's intrinsic rank tends to be higher, possibly even full rank as $r = \min(d, k)$. SVD is applied on the weight matrix W to obtain three matrices $U \in \mathbb{R}^{d\times d}$, $\Sigma \in \mathbb{R}^{d\times k}$, and $V \in \mathbb{R}^{k\times k}$ for low-rank approximation as

$$\mathrm{SVD}(W) = U\Sigma V^{\mathrm{T}} \tag{6}$$

where $U^{\mathrm{T}}U = \mathbb{I}^{d\times d}$, $VV^{\mathrm{T}} = \mathbb{I}^{k\times k}$, and Σ is a real diagonal matrix with no negative elements. In matrix Σ, the singular values are sequenced in descending order. To conduct low-rank r in the LoRA matrix, top-r singular values are reserved for presenting the most important parameter information, and matrices U, V are formalized to the corresponding dimension as

$$\begin{cases} U_r = U_{[:,:r]} \\ \Sigma_r = \Sigma_{[:r,:r]} \\ V_r^{\mathrm{T}} = V^{\mathrm{T}}{}_{[:r,:]} \end{cases} \tag{7}$$

where these three truncated low-rank matrices act as $U_r \in \mathbb{R}^{d\times r}$, $\Sigma_r \in \mathbb{R}^{r\times r}$, and $V_r^{\mathrm{T}} \in \mathbb{R}^{r\times k}$.

The low-rank approximation approach aims to approximate the parameter matrix W by the rank-r matrix $\bar{W}$ to minimize the euclidean distance between them. As matrix $W \in \mathbb{R}^{d\times k}$ and $r \leq \min(d, k)$, the constraints are formulated as below

$$\begin{cases} \|\bar{W} - W\|_\infty \leq \epsilon \\ \bar{W} = U_r \Sigma_r V_r^{\mathrm{T}} \end{cases} \tag{8}$$

where ϵ represents the low-rank approximation error. It is proven by [36] that when $r' \leq r \leq \mathrm{rank}(W) \leq \min(d, k)$

$$||W - \bar{W}||_\infty \leq ||W - W'||_\infty \tag{9}$$

which leads to a serious dilemma between compression effectiveness and approximation precision if the weight matrix W is close to full-rank, while the compression can be very efficient if there is a low intrinsic rank in matrix W.

Therefore, in order to analyze the relationship between the value of low-rank r, the effectiveness of low-rank decomposition, and low-rank approximation error, we introduce a compressed parameter ratio $p \in [0, 1]$. For a pretrained parameter matrix $W \in \mathbb{R}^{d\times k}$, its low-rank approximation approach introduces three matrices: $U_r \in \mathbb{R}^{d\times r}$, $\Sigma_r \in \mathbb{R}^{r\times r}$, and $V_r^T \in \mathbb{R}^{r\times k}$. After the matrix multiplication operation, it is possible to establish a correspondence between the two low-rank matrices $A \in \mathbb{R}^{r\times k}$ and $B \in \mathbb{R}^{d\times r}$ within the LoRA module

$$\begin{cases} \bar{B} = U_r \Sigma_r^{\frac{1}{2}} \\ \bar{A} = \Sigma_r^{\frac{1}{2}} V_r^{\mathrm{T}}. \end{cases} \tag{10}$$

Through this LoRA module, the number of trainable parameters is reduced from $d \times k$ to $d \times r + r \times k$ when $r \leq (d \times k)/(d + k)$. The compressed parameter ratio is defined as

$$p = \frac{r \times (d + k)}{d \times k}. \tag{11}$$

Furthermore, we investigate the correlation of the compressed parameter ratio p with the low-rank approximation error, which influences the selection of the rank value r on the LoRA module.

Specifically, if the low-rank approximation error consistently decreases as p increases, it implies a higher intrinsic dimensionality of the matrix. Consequently, opting for a smaller r may hurt the model's performance. Conversely, if the low-rank approximation error remains consistently low regardless of variations in p, it suggests the presence of a low intrinsic rank within the matrix. In such instances, employing a lower rank r within the LoRA module can achieve optimal fine-tuning performance while concurrently economizing on trainable parameters.

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According to the analyses of low-rank approximation, it is assumed that the parameter matrices of the multihead attention layers in the backbone of LoRA-Det are closed to full-rank while fully connected layers in the detection head have relatively low intrinsic ranks. Therefore, based on empirical trials and experiments, the rank r of the LoRA module is treated as a hyperparameter in LoRA-Det, set higher for the attention matrices in the backbone and lower for the linear layers in the detection head.

IV. Experiment Results and Analysis

A. Datasets

To assess the efficacy of the proposed method, three remote sensing image datasets for oriented object detection are utilized for validation.

1) *DOTAv1.0 [29]:* The DOTA dataset stands as a comprehensive collection for oriented object detection, encompassing 2806 aerial images and 188 282 labeled instances with oriented bounding box annotations across 15 different object categories. These categories include bridge (BR), harbor (HA), ship (SH), plane (PL), helicopter (HC), small vehicle (SV), large vehicle (LV), baseball diamond (BD), ground track field (GTF), tennis court (TC), basketball court (BC), soccer-ball field (SBF), roundabout (RA), swimming pool (SP), and storage tank (ST). The images are in high-resolution, ranging from 800 × 800 to 4000 × 4000 pixels. For the purpose of training, both training and validation subsets are utilized, while the test dataset is evaluated through the online submission of testing results to the DOTA evaluation server.
2) *HRSC2016 [30]:* The HRSC2016 dataset is utilized for the fine-grained ship detection within 31 categories annotated by oriented bounding boxes, comprising 1061 images whose sizes vary between 300 × 300 and 1500 × 900 pixels. The dataset is divided into a training set of 436 images and a validation set of 181 images, both of which are employed in the training process, with the remaining 444 images for testing purposes. To assess detection accuracy on HRSC2016, the evaluation leverages the mean average precision (mAP) metric, aligning with the evaluation standards set by PASCAL VOC 2007.
3) *DIOR-R [31]:* The DIOR-R dataset utilizes the images from the previously established DIOR dataset [37]. Oriented bounding boxes are annotated for each instance to facilitate tasks involving oriented object detection. The dataset encompasses 23 463 images and 192 518 instances, spanning a broad spectrum of scenes and 20 common object categories on remote sensing.

B. Data Configuration

To evaluate PEFT algorithms on new data, the experimental setup for the three datasets is as follows: The test set is used for validation, while both the training and validation sets are combined as training data. Half of the training data, sourced only from the original training set to prevent data leakage, serves as pretraining data to establish the model's lower performance bound. Full fine-tuning on the complete training and validation sets represents the upper bound. Our proposed method applies LoRA-based PEFT on the pretrained model using all training data and compares it with this full fine-tuning upper bound.

C. Parameter Settings

In our experiments, we utilize two NVIDIA RTX 3090, with a batch size of 4, employing the Adam optimizer. The learning rate is set to 0.0001, and weight decay is configured at 0.05. Our data augmentation strategy is limited on random flipping, and training is conducted using a single-scale approach. For the DOTA dataset, images are cropped into slices with the size of 1024 × 1024 pixels using a stride of 800 × 800, and the model is trained for 12 epochs. The HRSC2016 and DIOR-R datasets undergo image resizing to 1024 × 1024 and 800 × 800 pixels, respectively, before being fed into the network for training, with the HRSC2016 dataset undergoing 36 epochs of training and the DIOR-R dataset 12 epochs.

D. Baselines of Partial and Full Fine-Tuning

Models trained on 50% of the training data represent a lower bound for pretrained model performance, while full fine-tuning on the entire training set provides the upper bound. We expect LoRA-Det to exceed the lower bound in detection performance, closely approach the upper bound, and significantly reduce trainable parameters compared to full fine-tuning, demonstrating an effective tradeoff between performance and parameter efficiency.

E. Evaluation of Both Fine-Tuning Performance and Parameter Efficiency in LoRA-Det

The comprehensive experiments on three remote sensing datasets explore the efficiency of the proposed method, along with the comparison of various fine-tuning strategies, including: fine-tuning the backbone only; fine-tuning the detection head only; adopting LoRA modules in the backbone and full fine-tuning the whole detection head (referenced as LoRA in the table); the proposed LoRA-Det without full fine-tuning the convolutional layers of FPN and Oriented RPN modules, LoRA-Det with hybrid fine-tuning the FPN and Oriented RPN modules. The low-rank value r in the LoRA module of the proposed LoRA-Det is adjusted for optimizing model performance.

The detailed low-rank configurations of each LoRA module of the proposed LoRA-Det are listed in Table I, including every stage of the Swin transformer backbone and two shared fully connected network layers in the detection head. The original weight matrices dimensions d and k, the low-rank r value, and corresponding compressed parameter ratio p defined by (11) is presented to conclude the trainable parameter efficiency of LoRA-Det. As the network depth increases, closer to the end of the network architecture, the proportion of trainable parameters in the LoRA module rapidly decreases relative to the original weight parameters. This confirms

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TABLE I

LOW-RANK VALUE r CONFIGURATIONS AND THE COMPRESSED PARAMETER RATIOS OF LORA MODULES IN LORA-DET

LoRA Module	Backbone Stage 1	2	3	4	Detection head FC 1	FC 2
Matrix dimension d	96	192	384	768	1024	1024
Matrix dimension k	96	192	384	768	12544	1024
Low rank r	48	48	48	48	64	16
Compressed parameter ratio p	1	0.5	0.25	0.125	0.067602	0.03125

TABLE II

COMPARISON RESULTS OF THE PROPOSED METHOD AND OTHER FINE-TUNING METHODS ON DOTAV1.0 DATASET. EVALUATION BY MAP AND THE NUMBER OF TRAINABLE PARAMETERS

Method	Trainable Params(M)	Trainable Params Ratio(%)	mAP (AP50)
Pre-trained	44.76	100	73.74
Full fine-tune	44.76	100	77.84
Full fine-tune backbone only	30.25	67.5	50.20
Full fine-tune head only	17.26	38.5	67.80
LoRA	18.52	40.9	77.03
LoRA-Det	4.92	10.9	75.07
LoRA-Det-hybrid	5.52	12.3	76.27

TABLE III

COMPARISON RESULTS OF THE PROPOSED METHOD AND OTHER FINE-TUNING METHODS ON HRSC2016 DATASET. EVALUATION BY MAP AND THE NUMBER OF TRAINABLE PARAMETERS

Method	Trainable Params(M)	Trainable Params Ratio(%)	mAP (AP50)
Pre-trained	44.78	100	59.0
Full fine-tune	44.78	100	71.9
Full fine-tune backbone only	30.25	67.5	50.2
Full fine-tune head only	17.26	38.5	67.5
LoRA	18.53	40.9	69.3
LoRA-Det	4.92	10.9	68.3
LoRA-Det-hybrid	5.52	12.3	71.9

TABLE IV

COMPARISON RESULTS OF THE PROPOSED METHOD AND OTHER FINE-TUNING METHODS ON DIOR-R DATASET. EVALUATION BY MAP AND THE NUMBER OF TRAINABLE PARAMETERS

Method	Trainable Params(M)	Trainable Params Ratio(%)	mAP (AP50)
Pre-trained	44.77	100	59.3
Full fine-tune	44.77	100	66.4
Full fine-tune backbone only	30.25	67.5	33.2
Full fine-tune head only	17.26	38.5	48.7
LoRA	18.52	40.9	63.6
LoRA-Det	5.9	13.2	63.9
LoRA-Det-hybrid	6.49	14.5	64.2

the proposed LoRA-Det's highly efficient trainable parameter compression capability.

In LoRA-Det, the rank r is treated as a hyperparameter. For multihead attention layers in the backbone, r is set to 48, achieving the maximum compressed parameter ratio $p = 1$ in the first stage of the ViT backbone. For fully connected layers in the detection head, r is selected from lower values, specifically powers of 2 (e.g., 2, 4, 8, 16, 32, and 64). The final rank r is determined by balancing fine-tuning performance with the parameter budget constraints.

Through experimental analysis, the proposed method, under the condition of only using a minimal proportion of the model's parameters for LoRA-based PEFT and full fine-tuning for convolutional modules, significantly improves the pretrained model's generalization performance by adopting new data. LoRA-Det achieves to approximate of the performance of full model fine-tuning, and in some cases, it is on par with full fine-tuning upper bound.

On the DOTAv1.0 dataset, the performance of the pretrained model is measured as an mAP of 73.74%. Full fine-tuning of the model using 44.78 million parameters can elevate its performance to an upper bound of AP 77.84%. However, fine-tuning only the backbone or only the Oriented R-CNN head results in AP of 50.2% and 67.8%, respectively. Not only does this fail to enhance the model's detection capabilities, but it may also lead to model training collapse, deteriorating detection performance below the lower bound set by the pretrained model. While utilizing LoRA modules to fine-tune the backbone and full fine-tuning detection head can achieve satisfactory detection performance, it consumes 18.52 million trainable parameters, which is over three times the parameter quantity utilized by the proposed LoRA-Det.

In contrast, our proposed method, integrating LoRA for PEFT with full fine-tuning on convolutional layers, achieves an AP of 76.27% by training with just 5.52 million parameters, or 12.3% of the model's parameters. This equates to 97.98% of the model's performance upper bound. As illustrated in Table II, the model performance, computational requirements, and the volume of trainable parameters for various strategies are compared and listed.

For the HRSC2016 dataset, which features a smaller volume of data and imbalanced categories, conducting fine-grained object detection tasks on the 31 categories of ship targets presents significant challenges. The proposed method is capable of achieving performance nearly equivalent to that of full fine-tuning, as demonstrated in Table III. The performance of other strategies is inferior, consistent with the experimental results from the DOTAv1.0 dataset.

On the DIOR-R dataset, employing the proposed method to perform PEFT on the pretrained model with new data similarly yields significant benefits. A favorable balance between enhancing model performance and reducing the amount of parameters updated is achieved, consistent with the experimental conclusions drawn from the first two datasets. For this dataset, we select the different low-rank values r in the detection head, such as 128 and 64 for the first and second shared fully connected layers. As listed in Table IV, The combination of LoRA-based PEFT and full fine-tuning of certain convolutional modules of the detection algorithm reaches 96.75% of the full fine-tuning performance while using only 14.5% of the trainable parameters.

Table V presents the detailed results of mAP on every category on the DIOR-R dataset. Fig. 7 presents detection visualization of LoRA-Det and comparison with pretrained

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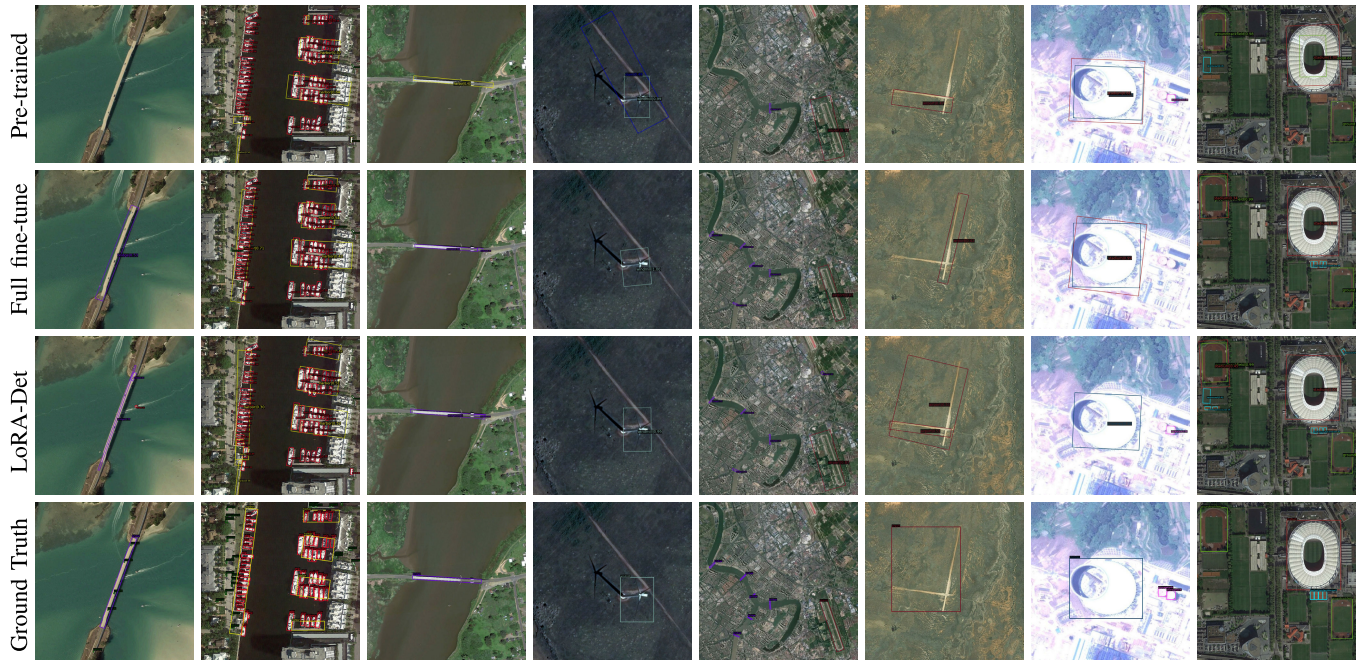


Fig. 7. Qualitative results of oriented object detection on DIOR-R testset. The compared visualizations between the proposed method and the pretrained model, full fine-tuned model, and ground truth. Diverse colors of bounding boxes represent different categories. The proposed method improves the issues of missing detection, false detection, and classification confusion in pretrained models.

TABLE V
COMPARISON WITH OTHER FINE-TUNING METHODS ON THE DIOR-R DATASET FOR EVERY CATEGORY

Methods	mAP	APL	APO	BF	BC	BR	CH	ETS	ESA	DAM	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM
Pre-trained	59.3	71.1	31.5	71.2	81.3	34.6	72.7	59.4	51.5	23.3	69.1	78.1	34.4	50.0	81.0	71.2	70.9	81.3	52.9	42.8	56.7
Full fine-tune	66.4	71.3	46.8	79.3	87.8	43.8	78.9	83.6	69.9	29.4	77.7	83.4	42.7	59.1	81.1	81.9	62.5	81.5	57.6	43.5	65.4
LoRA	63.6	71.9	36.9	78.5	86.3	39.0	72.6	78.2	66.0	25.3	68.9	83.0	34.4	53.0	81.1	81.2	70.7	81.4	52.0	46.6	64.6
LoRA-Det	63.9	71.5	39.2	78.6	81.4	39.0	72.7	76.0	64.9	25.6	73.2	82.5	39.5	53.9	81.1	80.7	71.1	81.5	54.1	47.4	64.3
LoRA-Det-hybrid	64.2	71.3	38.1	78.5	86.1	39.0	78.5	76.4	64.9	27.0	73.4	81.8	39.0	54.3	81.1	80.5	71.1	81.5	53.8	43.3	64.6

model, full fine-tuned model, and ground truth. LoRA-Det approaches the performance of fully fine-tuned models while also addressing issues of missing detection, false positives, and classification confusion in the pretrained model.

Fig. 8 illustrates the comparison of model performance and the number of trainable parameters between the proposed LoRA-Det and other fine-tuning approaches across three remote sensing image datasets. The horizontal axis represents the number of trainable parameters, while the vertical axis denotes the model's object detection performance, measured in mAP [38]. The size of the bubbles indicates the volume of trainable parameters; hence, bubbles located toward the upper left corner of the image, which are smaller in area, signify samples with superior overall performance.

The visualization results from experiments on the DOTAv1.0, HRSC2016, and DIOR-R datasets lead to a consistent conclusion: LoRA-Det, even with a significantly reduced amount of trainable parameters, still achieves competitive detection performance, managing to update far fewer parameters while approaching the performance of models fully fine-tuned. Similarly, the experiments validate that even with a larger number of trainable parameters, inappropriate fine-tuning methods fail to achieve optimal detection results.

F. Comparison With Lightweight State-of-the-Art Method

To validate the proposed algorithm, we compare its PEFT with the full fine-tuning of lightweight detection models under similar trainable parameter volumes, within satellite model update constraints. On the DOTAv1.0 dataset, the LoRA-Det model, fine-tuning only 12.4% of parameters, is evaluated against lightweight models like YOLOv8-OBB [39], PPYOLOE-R [40], and RTMDet [34], all with similar parameter counts. Training configurations on the single-scale level and data augmentation (random flipping) are consistent across models.

As shown in Table VI, the comparison of detection performance and trainable parameter volumes between LoRA-Det and other lightweight detectors demonstrates that LoRA-Det, with a similar volume of trainable parameters under the PEFT approach, surpasses the lightweight detection models RTMDet-tiny, and even exceeds the performance of RTMDet-small, which has a higher volume of trainable parameters. This suggests that the proposed LoRA-Det, leveraging a larger overall model parameter scale combined with PEFT technology, offers better model performance and generalizability compared to lightweight models. It is well-suited for

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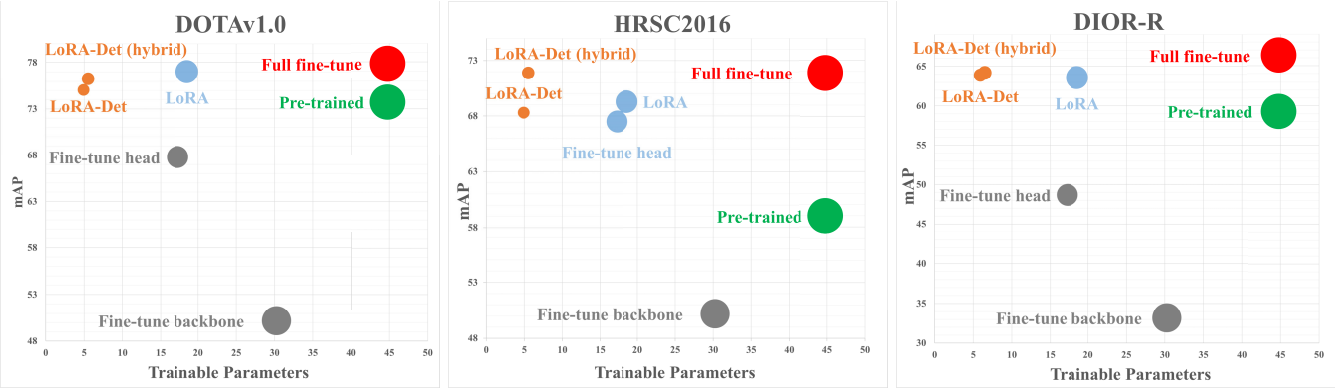


Fig. 8. Trainable parameter quantity versus oriented object detection performance mAP of the proposed LoRA-Det and other fine-tuning methods listed in the previous tables. Green bubbles represent the lower bound of pretrained models, red bubbles are the upper bound of full fine-tuning, and gray bubbles mean the performance is lower than pretrained models. The size of the bubbles indicates the volume of trainable parameters.

TABLE VI

COMPARISON OF LORA-DET WITH SEVERAL LIGHTWEIGHT DETECTORS (YOLOV8-OBB, PPYOLOE-R, AND RTMDET) ON THE NUMBER OF TRAINABLE PARAMETERS, AND MAP ACCURACY ON DOTAV1.0 TESTSET. THE AP OF EACH CATEGORY IS LISTED

Method	Trainable Params(M)	mAP	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
YOLOv8-obb-nano	3.08	73.57	87.67	77.44	49.83	61.85	73.34	77.65	88.11	90.47	80.16	87.50	59.68	63.13	74.72	75.59	56.34
PPYOLOE-R-small	8.24	73.82	88.80	79.24	45.92	66.88	80.41	82.95	88.20	90.61	82.91	86.37	55.80	64.11	65.09	79.50	50.43
RTMDet-tiny	4.88	73.88	88.63	76.14	48.99	66.82	80.57	80.66	88.28	90.83	85.00	87.01	54.81	62.44	68.15	75.63	54.30
RTMDet-small	8.86	75.24	88.71	79.89	50.485	69.82	80.19	81.73	88.68	90.86	82.78	87.44	55.38	65.12	72.80	80.56	54.22
LoRA-Det	4.92	75.07	89.38	79.56	51.59	72.10	77.48	83.13	87.80	90.88	84.52	85.42	60.44	65.35	66.24	67.28	64.82
LoRA-Det-hybrid	5.52	76.27	89.25	81.33	53.03	75.53	77.91	83.05	87.89	90.89	85.77	85.10	63.42	67.73	73.33	68.35	61.44

TABLE VII

COMPARISON RESULTS OF THE PROPOSED METHOD WITH PYRAMID VIT BACKBONE AND OTHER FINE-TUNING METHODS ON DIOR-R DATASET. EVALUATION BY MAP AND THE NUMBER OF TRAINABLE PARAMETERS

Methods PVT	Trainable Params(M)	Trainable Ratio(%)	mAP (AP50)	APL	APO	BF	BC	BR	CH	ETS	ESA	DAM	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM
Pre-trained	29.86	100	55.4	61.1	28.0	69.9	80.6	27.1	72.6	59.1	44.0	21.7	66.5	69.8	25.6	46.0	80.8	70.8	61.9	80.5	48.2	40.4	52.5
Full fine-tune	29.86	100	61.5	62.5	39.3	72.0	81.1	37.6	72.7	79.1	59.5	29.7	69.8	79.3	34.6	51.6	81.1	75.6	62.4	81.6	55.1	42.4	63.4
LoRA-Det	2.29	7.67	58.1	61.9	32.3	76.6	80.6	28.9	72.6	68.7	48.4	26.3	67.3	75.9	27.3	48.7	80.9	74.2	61.6	81.2	51.7	41.5	55.2
LoRA-Det-hybrid	5.51	18.45	60.3	62.1	35.0	77.4	81.0	33.4	72.6	75.6	56.7	25.8	68.3	77.8	31.7	50.3	81.0	76.5	61.7	81.4	53.5	42.0	62.5

deployment in satellite communication application scenarios. LoRA-Det presents great potential for improving the model performance of spaceborne detectors by PEFT on continuous data under the data update limitation and enough inference resources.

G. Generalizability and Effectiveness of LoRA-Det Across Different Backbones

To evaluate the generalizability of the proposed method for parameter efficient fine-tuning on various architectures of networks, the LoRA-Det is deployed on an alternative transformer-based backbone, the pyramid ViT (PVT) [41], with an Oriented R-CNN detection head. LoRA modules are applied in the W_q and W_v matrices of the multihead attention layers for each transformer stage, consistent with the original design. A rank r of 32 is assigned to the LoRA modules in the backbone, and ranks of 128 and 64 are used for the two fully connected layers in the detection head.

The total parameters for PVT-Oriented-RCNN amount to 29.86 M, while LoRA-Det with PVT-Oriented-RCNN includes 5.51 M trainable parameters, constituting 18.45% of the total model parameters. Table VII presents the comparison experiments of LoRA-Det with PVT-Oriented-RCNN through nonhybrid and hybrid strategies evaluated by AP50 metric. The proposed method achieves 98.05% of the performance of full fine-tuning with only 18.45% of the trainable parameters, similar to previous experiments. This result indicates that the proposed method is applicable and effective across different models with similar transformer-based architectures, demonstrating strong generalizability.

H. Analysis of Low-Rank Approximation and the Selection of Rank r Value of LoRA Module

To select an appropriate low-rank value r, it's essential to analyze if the pretrained model's parameter weight matrices have low intrinsic dimensions or are close to full rank. Using the proposed SVD-based low-rank approximation method,

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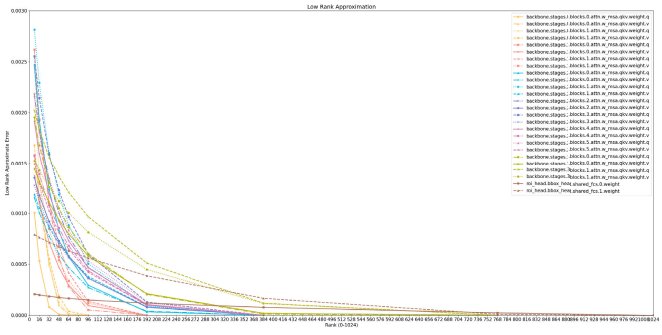


Fig. 9. Low-rank approximation results of each trainable parameter matrices in LoRA module of LoRA-Det.

we analyze LoRA-Det's trainable parameter matrices from the pretrained Oriented RCNN on the DOTAv1.0 dataset.

Each layer's parameter matrix has unique dimensions, as shown in Table I, so the maximum rank for low-rank approximation is set by the matrix's smallest dimension, $\text{rank}_{\max} = \min(d, k)$. For example, in the Swin transformer backbone stages 1 to 4, the maximum values for r are 96, 192, 384, and 768, corresponding to the query, key, and value matrices' dimensions in each stage's multihead attention. For the detection head's two shared fully connected layers, the maximum r is 1024, equal to $\min(1024, 12\,544)$.

Fig. 9 shows the low-rank approximation results for LoRA-Det's trainable parameter matrices. The backbone matrices exhibit relatively high approximation error, which gradually decreases as the rank r increases, indicating a large intrinsic dimension, nearly full rank. Conversely, the two shared fully connected layers in the detection head—despite their large dimensions ((1024, 12 544) and (1024, 1024))—maintain low approximation error even as r decreases, suggesting a small intrinsic dimension that allows for accurate low-rank approximation.

We analyze the relationship between low-rank approximation error and the compressed parameter ratio $p = (d \times r + r \times k)/(d \times k)$ to select appropriate rank values r for each LoRA-Det module. Fig. 10 shows the low-rank approximation results for stages 1–4 of the Swin transformer backbone. The x-axis represents p, where higher values indicate larger r and thus lower compression efficiency, while the y-axis shows approximation error. Results reveal that when p is low (indicating smaller r), approximation error is high, decreasing only as p approaches 1. This suggests that the backbone's parameter matrices lack extremely low intrinsic dimensions and cannot be accurately approximated with a low r. Therefore, a higher rank r is needed in the LoRA-Det backbone to maintain fine-tuning performance.

To determine an optimal r value, we conduct additional experiments by adjusting the r values in the LoRA modules of the backbone, varying trainable parameter counts, and evaluating fine-tuning performance on the DIOR-R dataset. Three settings are tested: Setting 1 uses the original LoRA-Det configuration; Setting 2 adjusts r in the backbone transformer blocks to 32; and Setting 3 applies a constant compressed parameter ratio across all backbone LoRA modules. Detection

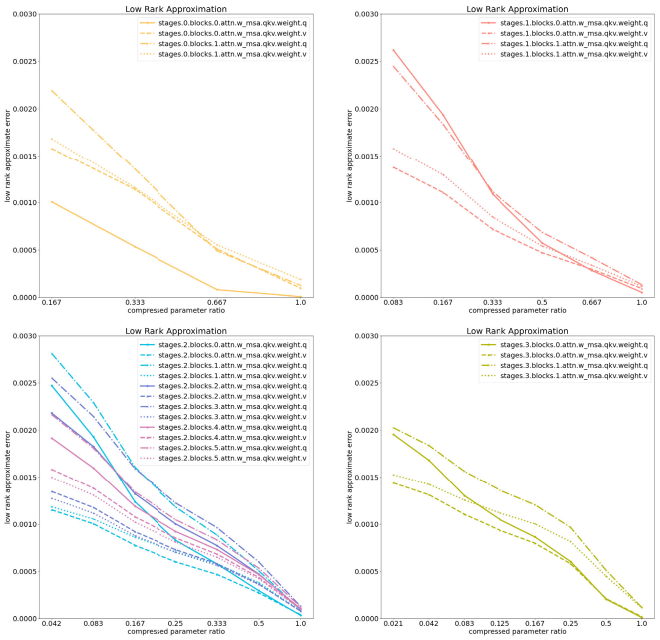


Fig. 10. Low-rank approximation of weight matrices on every stage on the Swin transformer backbone, stage 1, 2, 3, and 4 for left-top, right-top, left-bottom, and right-bottom subfigures.

TABLE VIII

LOW-RANK VALUE r CONFIGURATIONS AND COMPRESSED PARAMETER RATIOS OF LORA MODULES IN LORA-DET. THREE EXPERIMENTS WITH DIFFERENT SETTINGS ARE CONDUCTED IN DIOR-R DATASET. 'r' REPRESENTS LOW-RANK VALUE r IN LORA MODULE. 'p' REPRESENTS THE COMPRESSED PARAMETER RATIO

	LoRA Param.	Backbone Stage 1	2	3	4	Detection head FC 1	FC 2
Matrix Dimension	d	96	192	384	768	1024	1024
	k	96	192	384	768	12544	1024
Setting 1	r	48	48	48	48	64	16
	p	1	0.5	0.25	0.125	0.0676	0.03125
Setting 2	r	32	32	32	32	64	16
	p	0.667	0.333	0.167	0.083	0.0676	0.03125
Setting 3	r	24	48	96	192	64	16
	p	0.5	0.5	0.5	0.5	0.0676	0.03125

head parameters remain unchanged. Table VIII details the parameter settings for these comparison experiments.

Table IX summarizes the results of three experiments, each with different LoRA-Det configurations. Setting the rank $r = 32$ in the backbone results in 6.07 M trainable parameters and an AP50 of 64.0. In comparison, using $r = 48$ yields 6.49 M parameters and an AP50 of 64.2, indicating that increasing r improves performance within the current model size. With a uniformly compressed parameter ratio of 0.5, the model reaches 8.45 M parameters but only achieves a minor AP50 increase to 64.1. These findings suggest that increasing r does not consistently boost performance, and optimal tuning of r may depend on specific data distributions.

These results align with theoretical findings in [8], suggesting that larger model weight matrices in deep learning tend to exhibit low intrinsic dimensions. Analyzing the intrinsic

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TABLE IX
COMPARISON RESULTS OF THE PROPOSED METHOD WITH DIFFERENT PARAMETER SETTINGS OF RANK r OF LORA MODULE ON DIOR-R DATASET. EVALUATION BY MAP AND THE NUMBER OF TRAINABLE PARAMETERS. 'S1' REPRESENTS 'SETTING 1' IN TABLE VII

Method	Setting	Trainable Params(M)	Trainable Params Ratio(%)	mAP (AP50)
Pre-trained	-	44.77	100	59.3
Full fine-tune	-	44.77	100	66.4
LoRA	S1	18.52	40.9	63.6
LoRA-Det	S1	5.9	13.2	63.9
LoRA-Det-hybrid	S1	6.49	14.5	64.2
LoRA-Det-hybrid	S2	6.07	13.6	64.0
LoRA-Det-hybrid	S3	8.45	18.9	64.1

dimensions of pretrained weight matrices through low-rank approximation helps set the LoRA module's rank r in LoRA-Det, allowing an effective distribution of trainable parameters across the detection model to balance performance and parameter efficiency.

However, applying PEFT in normal-sized models is more challenging than in large models due to uncertain intrinsic dimensions in weight matrices. In LLMs, LoRA techniques typically set r to powers of 2 (e.g., $r = 2, 4, 8, 16$) due to their large parameter counts and low-rank weight matrices. In satellite-onboard applications, with fewer parameters and less consistent low-rank matrices, selecting r is more complex and often requires empirical tuning. Nearly full-rank parameter distributions in smaller models may lead to less effective PEFT, influenced by specific model frameworks. Consequently, designing fine-tuning strategies and architectures carefully is crucial, and developing methods to optimize r based on mathematical or statistical insights remains a valuable area for future research.

V. CONCLUSION

To address the need for updating limited quantities of parameters in the satellite onboard models for improving detection performance, this article presents a novel oriented object detection algorithm, LoRA-Det, which is based on hybrid PEFT technologies including LoRA with low-rank approximation and full fine-tuning modules. Employing the elaborate PEFT mechanisms of multiple dense layers in the whole model architectures, it achieves highly competitive performance under the constraint of limited parameter updates. Comprehensive experiments on three remote sensing image datasets validate the proposed method's efficiency. With only 12.4% of parameters updated, it achieves 98% of full fine-tuning performance. In addition, compared to lightweight detection networks with similar trainable parameter volumes, LoRA-Det optimally balances performance enhancement and parameter efficiency, leveraging continuous data to improve detection capabilities and generalization.

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